

ELLINGTON FIELD, TX, USA

WMO: 722436

Lat: 29.617N

Lon: 95.167W

Elev: 10

StdP: 101.21

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 12906

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.9	2.6	-7.5	2.0	5.5	-4.8	2.5	7.0	10.8	15.4	9.7	13.7	4.6	340	0.414

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.4	36.1	25.5	35.0	25.8	33.9	25.8	28.0	31.5	27.5	30.9	27.1	30.6	3.9	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.4	23.2	29.0	27.0	22.7	28.8	26.2	21.7	28.4	89.6	32.0	86.7	31.2	85.0	30.9	32.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.0	8.2	7.5	-1.7	38.2	0.9	2.2	-2.3	39.8	-2.9	41.0	-3.4	42.2	-4.1	43.8		
			-3.2	29.6	1.1	1.4	-4.0	30.5	-4.6	31.3	-5.2	32.1	-6.0	33.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	21.5	13.0	14.5	17.0	21.1	25.2	28.0	29.2	29.2	27.0	22.6	17.5	13.3
	DBStd	7.05	5.34	5.06	4.15	3.48	2.59	1.56	1.24	1.63	2.60	3.90	4.56	5.01
	HDD10.0	77	31	14	4	0	0	0	0	0	0	0	4	23
	HDD18.3	644	178	127	73	16	1	0	0	0	0	11	69	171
	CDD10.0	4278	123	140	222	333	472	541	594	596	512	392	228	124
	CDD18.3	1804	13	18	33	99	214	291	336	338	262	144	43	14
	CDH23.3	17823	41	67	139	578	1875	3182	3979	3939	2647	1096	237	44
	CDH26.7	7432	1	6	15	112	671	1396	1889	1884	1106	322	29	2
Wind	WSAvg	3.5	4.2	4.1	4.0	4.2	3.9	3.5	3.0	2.7	2.8	3.2	3.5	3.7
Precipitation	PrecAvg	1352	108	76	94	103	132	148	115	109	119	136	107	105
	PrecMax	2020	267	231	215	250	335	625	308	231	291	446	322	210
	PrecMin	602	15	13	20	2	1	7	35	13	41	1	18	28
	PrecStd	331	60	56	54	77	84	121	73	66	59	114	70	43
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.2	27.6	28.0	31.1	34.2	36.0	37.5	37.5	37.6	32.9	29.1	27.1
		MCWB	20.4	20.4	20.2	22.6	25.0	25.6	24.9	25.5	24.8	25.3	24.2	21.4
	2%	DB	24.3	25.3	26.9	29.1	32.5	34.1	35.9	36.2	34.8	31.3	27.5	25.0
		MCWB	20.0	20.1	20.4	22.1	24.5	25.6	25.5	25.9	25.4	24.4	21.9	20.9
	5%	DB	22.7	23.5	25.0	27.8	31.4	33.0	34.6	35.0	33.0	30.0	26.1	22.9
		MCWB	19.4	18.9	19.9	21.6	24.5	25.6	25.8	26.0	24.7	24.2	21.3	20.0
	10%	DB	21.3	22.0	23.1	27.0	30.2	32.5	33.7	33.8	32.2	28.0	24.8	21.8
		MCWB	18.9	18.6	19.3	21.3	24.2	25.6	25.9	26.1	24.7	23.3	20.6	19.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.5	22.0	23.4	25.1	27.1	28.0	28.5	28.8	28.7	27.5	25.3	23.0
		MCDB	24.2	25.1	26.3	28.8	30.3	31.5	31.9	32.5	32.1	29.4	27.6	24.8
	2%	WB	21.4	21.2	22.4	24.0	26.2	27.3	27.7	28.0	27.5	26.4	24.1	22.1
		MCDB	23.1	23.6	24.8	27.0	29.8	30.8	31.3	31.9	30.0	28.7	26.0	23.9
	5%	WB	20.7	20.5	21.6	23.4	25.6	26.8	27.2	27.5	26.9	25.7	23.0	21.1
		MCDB	22.2	22.5	23.6	26.0	29.3	30.4	31.0	31.1	29.6	28.1	25.1	22.8
	10%	WB	19.5	19.7	20.9	22.8	25.1	26.3	26.8	27.1	26.2	24.9	22.0	19.8
		MCDB	21.0	21.3	22.6	25.2	28.8	30.0	30.8	30.9	29.1	27.3	24.0	21.4
Mean Daily Temperature Range		MDBR	9.8	9.4	9.9	9.4	9.4	8.9	9.4	9.5	9.5	9.5	9.9	9.8
	5% DB	MCDBR	11.1	10.4	10.8	9.9	10.1	9.8	10.6	10.7	10.5	9.9	10.7	10.8
		MCWBR	7.7	6.0	6.1	4.7	4.1	3.4	3.5	3.3	3.7	4.5	6.7	7.3
	5% WB	MCDBR	9.6	8.4	8.7	7.9	8.9	8.9	9.4	9.2	8.5	7.9	9.0	9.6
		MCWBR	7.5	5.7	6.1	4.3	4.0	3.6	3.7	3.4	3.7	4.4	7.2	7.8
Clear-Sky Solar Irradiance	taub	0.343	0.362	0.376	0.439	0.481	0.473	0.499	0.486	0.458	0.382	0.372	0.353	
	taud	2.508	2.467	2.417	2.236	2.157	2.232	2.179	2.218	2.263	2.468	2.441	2.488	
	Ebn at Noon	892	904	911	859	821	823	800	809	820	871	849	861	
	Edh at Noon	90	102	114	141	153	142	149	142	131	101	96	88	
All-Sky Solar Radiation	RadAvg	2.89	3.26	4.47	5.33	5.96	6.18	5.97	5.72	4.97	4.35	3.29	2.56	
	RadStd	0.28	0.55	0.38	0.38	0.42	0.48	0.56	0.47	0.52	0.53	0.33	0.30	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (36 neighbors)	+0.33	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+111	+94

Nomenclature: See separate page